

Evan Maus

evan_maus@berkeley.edu | evwillow.com | linkedin.com/in/evwillow | github.com/evwillow | 740-971-8762 | San Francisco, CA

EDUCATION

University of California, Berkeley

Expected Dec 2026

Dual B.A. Economics & Data Science

Relevant Coursework: Advanced Econometrics (Econ 143); Probability for Data Science (Data 140); Introduction to Artificial Intelligence; Data, Inference & Decisions; Behavioral Finance; Asset Pricing & Portfolio Choice; Macroeconomic Policy; Linear Algebra.

SELECTED ACHIEVEMENTS

- Eagle Scout — Led 100 volunteers on a community trail expansion (Oct 2023)

EXPERIENCE

breakouts.trade

Feb 2025 – Present

Founder & Engineer

Quant Engine — breakouts-engine (Python)

- Built a systematic Qullamaggie channel-breakout engine, validated on a fresh 2024–2026 holdout touched exactly once — $R = 2.61$, CI [2.36–2.87] — behind a survivorship-free backtest of 66,753 trades (1998–2026): +3.18R mean per trade, 36.9% win rate.
- Designed a walk-forward, fractional-Kelly ML sizing model (refit yearly on prior data only) predicting per-trade expected R, gated by a QQQ-based market-regime filter.
- Built and deployed an automated IBKR trading robot (ib_async) — regime-gated, ML-sized entries on settled cash — running live on a real-money account since June 2026 as a systemd service with a kill switch.
- Built a nightly market-data pipeline (Polygon.io → S3 parquet lake) driving a daily breakout scanner.

Web Product — breakouts-trade (Next.js)

- Launched a Duolingo-style breakout-trading trainer + live scanner on the engine: 458 signups across 6 continents (~55 countries), 240 activated users, 10,203 practice drills.
- Shipped solo (~1,800 commits): Next.js 15 / React 19, Supabase/Postgres, Stripe, Google OAuth, Redis rate-limiting, and web-push/SMS/email alerts; Railway + VPS, CI/CD, Sentry.
- Designed one canonical rule/exit engine shared across the backtest, the drill, the live scanner, and the robot — so the setup a user practices is exactly the one they trade.
- Operate the system live end-to-end: broker integration, execution monitoring, and a reconciliation ledger audited against IBKR statements.

LLM Energy Benchmark

Aug 2025 – Present

Independent Researcher

- Built a benchmark testing whether prompt characteristics predict the energy cost of an LLM call: 68,407 client-side power-draw measurements across GPT-4o-mini, Llama-3.1-8B, and Mistral-Large via CodeCarbon — client-side draw during remote API calls, not provider-side inference energy.
- Engineered 52 linguistic/structural features over a 20,709-row subset of LMSYS-Chat-1M and built the full Python pipeline. The result was null — the strongest prompt-feature correlation was $r = 0.027$ — and I published the dataset and code at github.com/evwillow/llm-energy-benchmark with the null result intact.

Sustainable Delaware Ohio & N.C. Ohio Pollinator Pathway

Jan – Aug 2024

Full-Stack Developer

- Sole developer for two environmental nonprofits, leading technical decisions with non-technical stakeholders.
- Built and shipped each organization's website end-to-end — design, build, hosting, content — handing off maintainable sites to non-technical staff.

necoTECH

May – Sep 2023

Data & Product Intern

- Automated Python lead-tracking and data workflows; consolidated CRM, proposal, and marketing across three teams.
- Cleaned and structured lead and proposal data in Python to make downstream CRM and reporting workflows reliable.

Student Climate Action Team

Oct 2022 – Sep 2023

Founder & President

- Founded the organization and grew it to 20 active members through outreach and leadership.

TECHNICAL SKILLS

Python · TypeScript · SQL · Next.js · React · Node · PostgreSQL/Supabase · pandas · numpy · scikit-learn · statsmodels · SciPy · IBKR API (ib_async) · Stripe · PostHog · Docker · GitHub Actions · Redis · Tailwind · Railway/Vercel